

Integral Secant Distributions and Line-Code Obstructions for Complete (k, n) -Arcs

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Abstract

We study complete (k, n) -arcs through the numbers of maximal secants incident with each point. Splitting the exact block-pair count between the arc and its complement gives sharp integer bounds for both degree sequences; their real relaxation recovers the classical incidence bound. On every rational equality family with integral limiting degrees, the integer remainders give a positive linear improvement. For each factorization $\lambda = uv$, we make this improvement explicit when every external point lies on at least λ maximal secants. On characteristic-compatible families, near equality forces bounded modular repair, and completeness converts each repair point into further linear excess. In characteristic three, exact moment identities produce a small-weight word in the projective-plane line code. Its short line representation excludes a fixed linear neighborhood of the modular-repair bound and proves

$$t_{2q/3+1}(2, q) \geq \frac{q^2}{3} + \frac{5q}{3} - o(q), \quad q = 3^h.$$

At the exact linear endpoint, the three-line equality cases reduce to two triangular cores, and both are impossible for all sufficiently large fields. An analogous parity argument improves the characteristic-two bound under double maximal-secant coverage. The integer theorem also applies to selected blocks of symmetric designs and to minimal multiple blocking sets and nonextendible projective codes.

Keywords. complete (k, n) -arc; n -secant; multiple blocking set; symmetric design; projective code; incidence bound; modular stability.

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1 Introduction

A (k, n) -arc in a projective plane is a set of k points meeting every line in at most n points, with some line meeting it in exactly n points. It is complete if it is maximal under inclusion, or equivalently if every external point lies on an n -secant. The minimum size of a complete (k, n) -arc is denoted by $t_n(2, q)$ in $\text{PG}(2, q)$. Lower bounds for this quantity are usually obtained by counting n -secants or, dually, by bounding minimal multiple blocking sets; see Alabdullah–Hirschfeld [1] and Bishnoi–Mattheus–Schillewaert [8]. Recent work has also developed algebraic and curve-based methods for proving completeness itself; see, for example, Bastioni–Micheli [6], Korchmáros–Nagy–Szőnyi [11], and Bartoli–Timpanella [5].

The incidence estimate used in the existing lower bounds is an instance of the standard block-design incidence inequality; we use the normalization of Lund–Saraf [13] and the variance formulation of Murphy–Petridis [14]. Exact quotient equations for tactical decompositions are classical [7], but the selected family of all n -secants need not be tactical. Ramani’s recent first-incidence identity for uniform intersecting linear hypergraphs

[15] provides a related degree-defect viewpoint. Our argument uses both point classes and the intervals determined by their exact pair-count extrema. Congruence and stability methods for line-intersection distributions and small-weight line codes are developed by Szőnyi–Weiner and Adriaensen–Szőnyi–Weiner [18, 19, 2]. Here those methods enter only after the integer envelopes have bounded the number of exceptional dual lines.

This paper keeps the n -secants as a family rather than eliminating their number at the first counting step. The degrees of that family on the arc and on its complement have fixed sums, and their pair counts add to the number of pairs of secants. Integer convexity gives the exact extrema of each pair count, and the two ranges must overlap. Replacing the integer extrema by quadratic real averages recovers the classical incidence-mixing inequality. The pair identity and convex extrema are elementary; their simultaneous use keeps both the integer remainders and the secant number visible. On the rational equality families classified below, the remainders have linear order and the secant number localizes to a bounded interval. Modular stability and completeness then impose a second linear penalty.

The same integer remainder governs a different incidence structure in [17]: collisions between ordinary secants of an arc or cap, counted through the coplanar quadruples on each secant, rather than incidences between points and a selected family of maximal secants. That paper bounds complete caps in higher dimension and arcs complete outside a prescribed set of holes; this one bounds complete (k, n) -arcs in the plane. The counting step is shared, the degree sequences are not, and neither result contains the other.

The first application concerns ordinary completeness in characteristic three.

Theorem 1.1 (Characteristic-three bound). *For every $\varepsilon > 0$, there is h_0 such that, for $h \geq h_0$, every complete $(k, 2q/3 + 1)$ -arc in $\text{PG}(2, q)$, where $q = 3^h$, satisfies*

$$k \geq \frac{q^2}{3} + \left(\frac{5}{3} - \varepsilon\right)q. \quad (1.1)$$

Equivalently,

$$t_{2q/3+1}(2, q) \geq \frac{q^2}{3} + \frac{5q}{3} - o(q) \quad (q = 3^h \rightarrow \infty).$$

The incidence-mixing bound gives only $q^2/3 + q + O(1)$ in this regime. The integer degree bounds first raise the linear coefficient to $6/5$; a modular repair theorem of Szőnyi–Weiner [18] then raises it to $4/3$. The second step uses more than modular stability: each point changed in the dual repaired multiset becomes a primal line on which completeness forces a linear amount of external secant excess. At that center, exact moments produce a ternary line-code word of weight $O(q)$. Szőnyi–Weiner’s exact small-codeword theorem represents it on only a bounded number of lines. Two pointwise integer-shell inequalities force at least three generator lines and show that three lines require an additional $q/3 - o(q)$ points. Combining this gap with the modular-repair bound yields the coefficient $5/3$. At equality in the linear term, the signed three-line cores admit a finite structural reduction. Concurrent generators are impossible, and the two remaining triangular sign patterns each violate the arc cap; see Proposition 6.4. Thus the exact $5/3$ endpoint is absent, although this does not exclude positive sublinear displacement. Proposition 5.1 isolates the structural part for every characteristic-compatible rational equality family: at the leading equality density, the full dual secant set is a bounded edit of an exact $\lambda \pmod{p}$ multiset. This reduces, but does not solve, classification of the geometric extremizers.

The arithmetic source of the first improvement is visible whenever the real incidence bound is tight at a rational parameter. Fix a coverage multiplicity λ , factor it as $\lambda = uv$,

and put $d = u + v + 1$. For $h \in \mathbb{Z}$, define

$$c_{\text{bal}} = \frac{u(uv^2 + 4uv + 2u + 2v^2 + 4v + 1)}{2uv + u + v}, \quad (1.2)$$

$$h_* = -\frac{u(v+1)(u^2v + u^2 - v)}{2uv + u + v}, \quad (1.3)$$

$$\begin{aligned} \mu &= \frac{2v}{(u+v)(2uv + u + v + 1)}, \\ L(h) &= d - \frac{h}{u}, \quad C(h) = c_{\text{bal}} + \mu(h - h_*), \\ c_{\mathbb{Z}} &= \min_{h \in \mathbb{Z}} \max\{L(h), C(h)\}. \end{aligned} \quad (1.4)$$

Theorem 1.2 (Rational equality families). *Let $u, v \geq 1$, $\lambda = uv$, and $d = u + v + 1$. Suppose that a $(k, (u+1)m+1)$ -arc in a projective plane of order $q = dm$ has at least λ n -secants through every external point. Then*

$$k \geq udm^2 + c_{\mathbb{Z}}m - O_{u,v}(1). \quad (1.5)$$

Moreover, the classical incidence-mixing bound has linear coefficient

$$c_{\text{mix}} = \frac{u(uv + 3u + 2v + 3)}{2u + 1},$$

and

$$c_{\text{bal}} - c_{\text{mix}} = \frac{u(v+1)(u^2 + u + 1)}{(2u+1)(2uv + u + v)} > 0. \quad (1.6)$$

Every rational equality parameter of the continuous bound whose limiting average internal degree is an integer $a > \lambda$ arises from exactly one ordered factorization $\lambda = uv$.

Thus the strict improvement is not an isolated rounding effect. At fixed coverage multiplicity there is one equality family for each divisor, and every family has a positive linear integrality penalty. The two integers adjacent to h_* determine $c_{\mathbb{Z}}$, so the coefficient is explicit.

The same integer necessary condition holds in arbitrary symmetric designs. In a projective plane there are two further bounds. At most $\lfloor (k-1)/(n-1) \rfloor$ n -secants pass through a point of the arc, and at most $\lfloor k/n \rfloor$ pass through an external point. These geometric bounds sharpen the general condition for complete arcs.

In characteristic two the first family in which the two first-order bounds are simultaneously tight yields a second application.

Theorem 1.3 (Characteristic-two parity bound). *For every $\varepsilon > 0$, there is h_0 such that, for $h \geq h_0$, if $q = 2^h$ and every external point of a $(k, 3q/4 + 1)$ -arc in $\text{PG}(2, q)$ lies on at least two $(3q/4 + 1)$ -secants, then*

$$k \geq \frac{q^2}{2} + \left(\frac{73}{48} - \varepsilon \right) q. \quad (1.7)$$

The proof separates odd and even repair support. The parity of the selected secant family fixes the parity of the support, while the symmetric difference of its generator lines supplies respectively $q/4 - O(1)$ or $q/2 - O(1)$ external points of excess degree.

The projective-code translation is immediate. A spanning point set in $\text{PG}(r-1, q)$ gives a projective $[k, r, k-n]_q$ code. Hyperplanes meeting the point set in n columns correspond to minimum-weight codewords, and a candidate extension column is obstructed by each such

hyperplane through it. The same integer degree theorem therefore bounds nonextendibility with multiple minimum-weight witnesses. We keep the finite-geometry language primary and record this translation after the planar applications.

The paper is organized as follows. Section 2 proves the symmetric-design inequality and identifies incidence mixing as its real relaxation. The next two sections specialize to n -secants and prove Theorem 1.2. The modular-stability section proves an intermediate characteristic-three bound, the line-code section proves Theorem 1.1, and the next section proves Theorem 1.3. The final section gives the blocking-set and coding translations and records the exact computational checks used to guard the finite algebra.

2 Sharp integer degree bounds

For integers $N \geq 1$ and $R \geq 0$, write $R = aN + w$ with $0 \leq w < N$, and define

$$\Phi_{\min}(N, R) = (N - w) \binom{a}{2} + w \binom{a+1}{2}. \quad (2.1)$$

If $L \leq U$, $NL \leq R \leq NU$, put $E = R - NL = t(U - L) + w$, where $0 \leq w < U - L$, and define

$$\Phi_{\max}(N, L, U, R) = N \binom{L}{2} + LE + t \binom{U-L}{2} + \binom{w}{2}. \quad (2.2)$$

When $L = U$, the latter expression has its evident constant interpretation.

We will use the following exact form of the balancing loss. If $R = Na + w$, then every integer sequence with sum R satisfies

$$\sum_{i=1}^N \binom{z_i}{2} - \Phi_{\min}(N, R) = \frac{1}{2} \sum_{i=1}^N (z_i - a)(z_i - a - 1). \quad (2.3)$$

Every summand on the right is nonnegative, and every index with $z_i \notin \{a, a+1\}$ contributes at least one to the left side. Thus an $O(m)$ total balancing loss permits only $O(m)$ departures from the two adjacent balanced degrees. This quantitative consequence, rather than an unstated stability principle, will control the exceptional lines in Sections 5 and 7.

Lemma 2.1 (Balanced and filled degree sequences). *Among integer sequences $z_1, \dots, z_N$ with sum R , the minimum of $\sum_i \binom{z_i}{2}$ is $\Phi_{\min}(N, R)$. With the additional bounds $L \leq z_i \leq U$, its maximum is $\Phi_{\max}(N, L, U, R)$.*

Proof. If $z_i \geq z_j + 2$, replacing (z_i, z_j) by $(z_i - 1, z_j + 1)$ lowers the sum by $z_i - z_j - 1$. Repetition leaves only the adjacent values $a, a+1$, proving (2.1). For the maximum, the reverse exchange moves one unit from a smaller nonminimal coordinate to a larger nonmaximal one and raises the sum. Hence all but at most one coordinate lie at L or U ; counting the filled coordinates gives (2.2). $\square$

Theorem 2.2 (Exact integer necessary condition). *Let $\mathcal{D}$ be a symmetric 2 -(v, K, Λ) design whose point set is partitioned into $A \sqcup B$. Let $\mathcal{S}$ be a family of T blocks, each meeting A in s points. Suppose the selected-block degrees satisfy integer bounds*

$$L_x \leq z_x \leq U_x \quad (x \in A \sqcup B).$$

At the fixed totals

$$\sum_{x \in A} z_x = sT, \quad \sum_{x \in B} z_x = (K - s)T,$$

let $[P_A^-, P_A^+]$ and $[P_B^-, P_B^+]$ be the exact minimum and maximum of the two sums $\sum \binom{z_x}{2}$ over their respective integer boxes. Then

$$[P_B^-, P_B^+] \cap \left[\Lambda \binom{T}{2} - P_A^+, \Lambda \binom{T}{2} - P_A^- \right] \neq \emptyset. \quad (2.4)$$

For uniform boxes the endpoints are given by Lemma 2.1.

Proof. Count triples (x, S_1, S_2) with $S_1, S_2 \in \mathcal{S}$, $S_1 \neq S_2$, and $x \in S_1 \cap S_2$. Two blocks of a symmetric design meet in exactly Λ points, so

$$\sum_{x \in A} \binom{z_x}{2} + \sum_{x \in B} \binom{z_x}{2} = \Lambda \binom{T}{2}. \quad (2.5)$$

Each sum lies in its exact integer interval. Reflecting the interval for A through the right side of (2.5) gives (2.4). $\square$

The interval overlap is necessary, not sufficient: membership of an integer in the two enclosing intervals does not assert that a selected block family realizes the corresponding pair counts.

The real relaxation is familiar, but the following formulation makes the loss of information exact.

Corollary 2.3 (Incidence mixing as the real relaxation). *In the setting of Theorem 2.2, assume additionally that A and B are nonempty and that $T > 0$. Then*

$$T \left(\frac{s^2}{|A|} + \frac{(K-s)^2}{|B|} - \Lambda \right) \leq K - \Lambda. \quad (2.6)$$

This is algebraically identical to the incidence-graph mixing inequality for the point set B and block family $\mathcal{S}$. Equality requires constant selected-block degree on both point classes.

Proof. Cauchy–Schwarz and (2.5) give

$$\frac{s^2 T^2}{|A|} - sT + \frac{(K-s)^2 T^2}{|B|} - (K-s)T \leq \Lambda T(T-1).$$

Divide by T and use $K = s + (K-s)$. Conversely, expanding the standard mixing inequality and substituting $K(K-1) = \Lambda(v-1)$ gives the same expression. Equality in the two Cauchy inequalities is equivalent to constant degrees. $\square$

For uniform lower bounds without an upper cap, the balancing remainder has the closed form

$$\Phi_{\min}(N, R) = \frac{R^2/N - R}{2} + \frac{w(N-w)}{2N}, \quad R \equiv w \pmod{N}, \quad 0 \leq w < N. \quad (2.7)$$

The last term is precisely what the real relaxation discards. Along some rational equality families, it has linear size and cannot be absorbed into lower-order error.

3 The distribution of n -secants

Let $\mathcal{K}$ be a (k, n) -arc in a projective plane of order q . We use the standard secant notation

$$\tau_i = \#\{i\text{-secants of } \mathcal{K}\},$$

and write $\rho_i(P)$ for the number of i -secants through $P \in \mathcal{K}$, and $\sigma_i(Q)$ for the number through $Q \notin \mathcal{K}$. Only $i = n$ will occur below. Put

$$b = q^2 + q + 1 - k, \quad \tau = q + 1 - n, \quad T = \tau_n.$$

The geometry supplies two bounds that need not hold for a selected block family in an arbitrary design.

Lemma 3.1 (Secants in a pencil). *For every $P \in \mathcal{K}$ and $Q \notin \mathcal{K}$,*

$$\rho_n(P) \leq D := \left\lfloor \frac{k-1}{n-1} \right\rfloor, \quad \sigma_n(Q) \leq M := \left\lfloor \frac{k}{n} \right\rfloor. \quad (3.1)$$

Moreover,

$$\sum_{P \in \mathcal{K}} \rho_n(P) = nT, \quad \sum_{Q \notin \mathcal{K}} \sigma_n(Q) = \tau T, \quad (3.2)$$

$$\sum_{P \in \mathcal{K}} \binom{\rho_n(P)}{2} + \sum_{Q \notin \mathcal{K}} \binom{\sigma_n(Q)}{2} = \binom{T}{2}. \quad (3.3)$$

Proof. The n -secants through P have disjoint sets of $n-1$ further points of $\mathcal{K}$. Those through Q have disjoint n -point intersections with $\mathcal{K}$. This proves (3.1). The first two equations count incidences. Every two distinct projective lines meet once, either in $\mathcal{K}$ or outside it, which gives (3.3). $\square$

We will sometimes impose the stronger hypothesis

$$\sigma_n(Q) \geq \lambda \quad (Q \notin \mathcal{K}), \quad (3.4)$$

where $\lambda \geq 1$. For $\lambda = 1$, this is precisely completeness. Rather than introduce another name for (3.4), we state the multiplicity explicitly in each result.

Corollary 3.2 (Integer feasibility for the n -secants). *Assume (3.4). Then there is an integer T satisfying*

$$\left\lceil \frac{\lambda b}{\tau} \right\rceil \leq T \leq \left\lfloor \frac{k(k-1)}{n(n-1)} \right\rfloor, \quad (3.5)$$

$$[\Phi_{\min}(b, \tau T), \Phi_{\max}(b, \lambda, M, \tau T)] \cap \left[\binom{T}{2} - \Phi_{\max}(k, 0, D, nT), \binom{T}{2} - \Phi_{\min}(k, nT) \right] \neq \emptyset. \quad (3.6)$$

Proof. Equation (3.4) and (3.2) give the lower bound for T . Each n -secant contains $\binom{n}{2}$ pairs of points of $\mathcal{K}$, and each pair lies on one line, giving the upper bound. Apply Theorem 2.2 to the point-line design of the plane, using Lemma 3.1 for the degree bounds. $\square$

If

$$v_K \equiv nT \pmod{k}, \quad 0 \leq v_K < k, \quad v_{\text{ext}} \equiv \tau T \pmod{b}, \quad 0 \leq v_{\text{ext}} < b,$$

then the two minimum terms in (3.6) imply

$$T \left(\frac{n^2}{k} + \frac{\tau^2}{b} - 1 \right) + \frac{v_K(k - v_K)}{kT} + \frac{v_{\text{ext}}(b - v_{\text{ext}})}{bT} \leq q. \quad (3.7)$$

Discarding the two nonnegative residue terms gives the familiar real incidence bound. The coefficient of T is strictly positive, since

$$\frac{n^2}{k} + \frac{\tau^2}{b} \geq \frac{(n + \tau)^2}{k + b} = \frac{(q + 1)^2}{q^2 + q + 1} > 1.$$

We may therefore use $T \geq \lambda b / \tau$ without reversing the inequality, and obtain

$$\frac{\lambda b}{\tau} \left(\frac{n^2}{k} + \frac{\tau^2}{b} - 1 \right) \leq q. \quad (3.8)$$

After denominators are cleared, (3.8) is the quadratic in Bishnoi–Mattheus–Schillewaert [8, Theorem 8.1]. Thus the classical bound is the real relaxation of (3.6); the residue terms record part of the discarded integer information. Schillewaert [16] determines the equality cases of the corresponding bound for minimal multiple blocking sets.

Remark 3.3. For a complete (k, n) -arc one takes $\lambda = 1$. The stronger values of λ below mean only that every external point is on at least λ n -secants. This is the same multiplicity hypothesis used in the dual form of Bishnoi–Mattheus–Schillewaert [8, Theorem 8.1].

4 Rational equality parameters

We first determine when the real incidence bound can be tight while the limiting average internal n -secant degree is an integer. Suppose that

$$\frac{n}{q} \longrightarrow \alpha, \quad \frac{k}{q^2} \longrightarrow \beta,$$

that equality holds asymptotically in (3.8), and that the limiting average internal degree is an integer $a > \lambda$. Equality in the incidence and coverage bounds gives

$$a = \frac{\lambda\alpha(1 - \beta)}{(1 - \alpha)\beta}, \quad \lambda(\alpha - \beta)^2 = \beta(1 - \alpha). \quad (4.1)$$

If $d = a - \lambda$, eliminating β gives

$$d^2\alpha^2 - d(d + 1)\alpha + a = 0, \quad (d + 1)^2 - 4a = (a - \lambda - 1)^2 - 4\lambda. \quad (4.2)$$

Lemma 4.1 (Classification of the rational parameters). *Fix an integer $\lambda \geq 1$. The triples $(\alpha, \beta, a) \in \mathbb{Q}^2 \times \mathbb{Z}$ satisfying (4.1),*

$$0 < \beta < \alpha < 1, \quad a > \lambda,$$

are in bijection with ordered factorizations $\lambda = uv$ in positive integers. They are

$$d = u + v + 1, \quad \alpha = \frac{u + 1}{u + v + 1}, \quad \beta = \frac{u}{u + v + 1}, \quad a = (u + 1)(v + 1). \quad (4.3)$$

Proof. The last expression in (4.2) must be an integer square. Factoring the difference of squares gives two positive factors of 4λ with the same parity. Both factors are even; write them as $2u$ and $2v$. Then $uv = \lambda$, and substitution gives (4.3). Conversely, the displayed values satisfy (4.1). The inequalities on α and β make the factors positive, and their order distinguishes the two roots, so the correspondence is one-to-one. $\square$

We next control $T = \tau_n$ near one of these parameters. The argument is stated separately because it is also used in the modular refinements.

Lemma 4.2 (Number of n -secants near equality). *Fix $u, v \geq 1$ and constants $C, K \geq 0$, and put $d = u + v + 1$. Suppose*

$$q = dm, \quad n = (u + 1)m + 1, \quad k = udm^2 + c_m m + e_m,$$

where $|c_m| \leq C$ and $|e_m| \leq K$. If every external point is on at least uv n -secants and

$$F(T) := \Phi_{\min}(k, nT) + \Phi_{\min}(q^2 + q + 1 - k, vmT) - \binom{T}{2} \quad (4.4)$$

is nonpositive, then

$$T = ud(v + 1)m + O_{u,v,C,K}(1). \quad (4.5)$$

Proof. Set

$$a = (u + 1)(v + 1), \quad \lambda = uv, \quad T_* = ud(v + 1)m,$$

and let $b = q^2 + q + 1 - k$. Thus

$$b = d(v + 1)m^2 + (d - c_m)m + 1 - e_m.$$

The external incidence count gives

$$T \geq T_0 := \left\lceil \frac{uvb}{vm} \right\rceil = T_* + r_m, \quad (4.6)$$

where $|r_m| \leq R_0$ for a constant $R_0 = R_0(u, v, C, K)$.

We first record a uniform bound at this left endpoint. Since

$$nT_0 - ak = O_{u,v,C,K}(m), \quad 0 \leq vmT_0 - \lambda b = O_{u,v,C,K}(m),$$

the balanced internal degrees at T_0 lie in $\{a - 1, a, a + 1\}$, and the balanced external degrees lie in $\{\lambda, \lambda + 1\}$, for all sufficiently large m . Formula (2.1) therefore gives

$$\Phi_{\min}(k, nT_0) = k \binom{a}{2} + O_{u,v,C,K}(m), \quad \Phi_{\min}(b, vmT_0) = b \binom{\lambda}{2} + O_{u,v,C,K}(m).$$

After substituting $k = udm^2 + O(m)$, $b = d(v + 1)m^2 + O(m)$, and $T_0 = T_* + O(1)$, the coefficient of m^2 in

$$k \binom{a}{2} + b \binom{\lambda}{2} - \binom{T_0}{2}$$

vanishes by the identity

$$ud \binom{a}{2} + d(v + 1) \binom{\lambda}{2} = \frac{u^2 d^2 (v + 1)^2}{2}.$$

Consequently there is a constant $A = A(u, v, C, K)$ such that

$$|F(T_0)| \leq Am \quad (4.7)$$

for all sufficiently large m .

Write $\Delta F(T) = F(T+1) - F(T)$. Its exact value is

$$\Delta F(T) = \sum_{i=0}^{n-1} \left\lfloor \frac{nT+i}{k} \right\rfloor + \sum_{i=0}^{vm-1} \left\lfloor \frac{vmT+i}{b} \right\rfloor - T. \quad (4.8)$$

For every $T \geq T_0$, the first family of floors is at least $a-1$, and the second is at least λ . Hence, uniformly for $0 \leq s \leq vm$,

$$\Delta F(T_0 + s) \geq n(a-1) + vm\lambda - T_0 - s = vm - s + O_{u,v,C,K}(1). \quad (4.9)$$

Choose a fixed integer $S > 2A/v + 2$. Equations (4.7) and (4.9) show that $F(T_0 + s) > 0$ for $S \leq s \leq vm/2$, once m is sufficiently large. From $vm/2$ to vm , the same lower bound is bounded below by a constant, while the accumulated increase at $vm/2$ has order m^2 . Thus

$$F(T_0 + s) > 0 \quad (S \leq s \leq vm). \quad (4.10)$$

It remains to exclude a later return to nonpositive values. Using $\lfloor x \rfloor \geq x - 1$ in (4.8) gives

$$\Delta F(T) \geq T \left(\frac{n^2}{k} + \frac{v^2 m^2}{b} - 1 \right) - (n + vm). \quad (4.11)$$

Here

$$\frac{n^2}{k} + \frac{v^2 m^2}{b} - 1 = \frac{1}{u(v+1)} + O(m^{-1}).$$

At $T = T_0 + vm$, the right side of (4.11), divided by m , tends to

$$\frac{ud(v+1) + v}{u(v+1)} - d = \frac{v}{u(v+1)} > 0.$$

The coefficient of T in (4.11) is also positive for all sufficiently large m . Therefore $\Delta F(T) > 0$ for every $T \geq T_0 + vm$. Together with (4.10), this shows that $F(T) \leq 0$ forces $0 \leq T - T_0 < S$. Since $T_0 - T_* = O_{u,v,C,K}(1)$, this proves (4.5). $\square$

Proof of Theorem 1.2. Lemma 4.1 proves the classification. If $c_m = (k - udm^2)/m$ exceeds a fixed constant larger than $c_{\mathbb{Z}}$, the claimed bound is automatic. On the other hand, expanding the classical bound (3.8) at these parameters gives $c_m \geq c_{\text{mix}} - O_{u,v}(m^{-1})$. Thus c_m is also bounded below, and in the remaining range Lemma 4.2 applies with $e_m = 0$, uniformly to the actual number of n -secants, because (3.6) implies $F(T) \leq 0$. Hence

$$T = ud(v+1)m + h$$

for an integer h in a fixed finite set. Substitution in the external coverage inequality $uvb \leq vmT$ gives

$$k \geq udm^2 + L(h)m - O_{u,v}(1). \quad (4.12)$$

Write $a = (u+1)(v+1)$. The balanced external degrees are uv and $uv+1$. On the branch $nT \leq ak$, the balanced internal degrees are $a-1$ and a , and direct substitution in (2.1) gives the affine expression below. On the other branch they are a and $a+1$. This branch change replaces only the internal term in F . The function $\Phi_{\min}(k, nT)$ is convex

in T , so its high-branch value lies above the affine continuation of its low-branch formula. Adding the same external term and $-(\binom{T}{2})$ preserves this pointwise inequality. Consequently

$$\frac{F(T)}{m} \geq \frac{(u+v)(2uv+u+v+1)}{2}(C(h)-c_m) - O_{u,v}(m^{-1}). \quad (4.13)$$

The integer condition (3.6) implies $F(T) \leq 0$, so

$$k \geq udm^2 + C(h)m - O_{u,v}(1).$$

The affine functions L and C are decreasing and increasing, respectively, and meet at h_* . Their maximum over the integers is minimized at $\lfloor h_* \rfloor$ or $\lceil h_* \rceil$. Since h ranges over a fixed finite set, the bounded terms are uniform. This proves (1.5).

Expanding the real bound (3.8) gives c_{mix} . Subtraction from c_{bal} gives the factorization (1.6). $\square$

For the first three ordered factorizations, the integer coefficients are

$$(u, v) = (1, 1) : \frac{18}{5}, \quad (u, v) = (1, 2) : \frac{9}{2}, \quad (u, v) = (2, 1) : 6. \quad (4.14)$$

The first is the ordinary-completeness family $q = 3m$, $n = 2m + 1$. The other two have two n -secants through every external point and limiting ratios $n/q = 1/2$ and $3/4$.

There is also an elementary description of the parameters for which the coverage and pair-count inequalities are simultaneously tight to first order.

Proposition 4.3 (Simultaneous first-order equality). *Put $M_0 = 2uv + u + v$. Then h_* is an integer if and only if*

$$M_0 \mid u^2(u+1)(u^2+u+1). \quad (4.15)$$

In particular, the case $v = u$ has simultaneous first-order equality for every even u .

Proof. Since $M_0 = (2u+1)v + u$, $\gcd(M_0, 2u+1) = \gcd(u, 2u+1) = 1$. Modulo M_0 ,

$$(2u+1)(v+1) \equiv u+1, \quad (2u+1)(u^2v+u^2-v) \equiv u(u^2+u+1).$$

It follows that

$$(2u+1)^2 u(v+1)(u^2v+u^2-v) \equiv u^2(u+1)(u^2+u+1) \pmod{M_0}.$$

Because $2u+1$ is invertible modulo M_0 , integrality of h_* is equivalent to (4.15). When $v = u$,

$$h_* = -\frac{u(u^2+u-1)}{2},$$

which is integral for even u . $\square$

5 Modular stability and completeness

Assume now that the plane is $\text{PG}(2, q)$, where $q = p^e$. Dualizing the set of n -secants gives a point set $\mathcal{L}^*$ in the dual plane. The intersection of $\mathcal{L}^*$ with the line dual to a point X has size $\rho_n(X)$ or $\sigma_n(X)$, according as $X \in \mathcal{K}$ or $X \notin \mathcal{K}$. At a rational equality parameter from Section 4, the two limiting intersection numbers are

$$\lambda = uv, \quad a = (u+1)(v+1), \quad a - \lambda = u + v + 1.$$

When $u + v + 1$ is a power of p , these two numbers are congruent modulo p .

The two characteristic-specific arguments follow the same chain. A bounded first-order coefficient localizes T to within $O(1)$ of its continuous value, and the exact balancing loss then leaves only $O(q)$ wrong-residue dual lines. Modular stability compresses those lines to $O(1)$ repair support; dualizing this support gives primal lines on which completeness forces a linear external excess.

Szőnyi and Weiner prove that, for $q = p^e > 27$ with $e > 1$, a multiset in $\text{PG}(2, q)$ having δ lines with the wrong intersection residue can, under their stated upper bound on δ , be changed at exactly $\lceil \delta/(q+1) \rceil$ distinct support points into a multiset having the prescribed residue on every line [18, Theorem 1.2]. In our application $e > 2$, $\delta = O(q)$, and their threshold is $(\lfloor \sqrt{q} \rfloor + 1)(q + 1 - \lfloor \sqrt{q} \rfloor)$. Thus the threshold holds for all sufficiently large q , and the number of changed support points is bounded. Related applications of modular stability to arc and direction problems appear in Csajbók–Weiner [10] and Adriaensen–Szőnyi–Weiner [2].

The same argument gives a structural statement on every characteristic-compatible rational equality family.

Proposition 5.1 (Stability near a rational equality family). *Fix $u, v \geq 1$, put $\lambda = uv$, and suppose $d = u + v + 1 = p^j$ for some prime p . Along a family*

$$q = p^e = dm, \quad n = (u + 1)m + 1, \quad k = udm^2 + O(m), \quad e \longrightarrow \infty,$$

suppose every external point of a (k, n) -arc lies on at least λ n -secants. Let $\mathcal{L}^$ be the dual point set of all n -secants. Then only $O(q)$ lines of the dual plane meet $\mathcal{L}^*$ in a number of points different from $\lambda \pmod{p}$. Moreover, $\mathcal{L}^*$ can be changed at $O(1)$ distinct support points into a multiset meeting every line in $\lambda \pmod{p}$ points.*

Proof. Write $k = udm^2 + c_m m$, where c_m is bounded. By Corollary 3.2, the actual secant number satisfies $F(T) \leq 0$. Lemma 4.2 therefore gives

$$T = ud(v + 1)m + O(1).$$

Put $a = (u + 1)(v + 1)$, so $a - \lambda = d \equiv 0 \pmod{p}$. The exact formulas defining F , evaluated with bounded c_m and this value of T , give $F(T) = O(m)$. The pair identity therefore gives

$$\begin{aligned} -F(T) &= \left(\sum_{P \in \mathcal{K}} \binom{\rho_n(P)}{2} - \Phi_{\min}(k, nT) \right) \\ &\quad + \left(\sum_{Q \notin \mathcal{K}} \binom{\sigma_n(Q)}{2} - \Phi_{\min}(b, (q + 1 - n)T) \right) = O(m). \end{aligned}$$

Both parentheses are nonnegative. For all sufficiently large m , put $b_0 = \lfloor nT/k \rfloor \in \{a - 1, a\}$. For every integer $z \notin \{b_0, b_0 + 1\}$,

$$|z - a| \leq (z - b_0)(z - b_0 - 1).$$

Equation (2.3) therefore shows that all but $O(m)$ internal degrees lie in the adjacent balanced pair $\{b_0, b_0 + 1\} \subseteq \{a - 1, a, a + 1\}$, and that

$$\sum_{\substack{P \in \mathcal{K} \\ \rho_n(P) \notin \{b_0, b_0 + 1\}}} |\rho_n(P) - a| = O(m).$$

Since

$$ak - nT = \sum_{P \in \mathcal{K}} (a - \rho_n(P)) = O(m),$$

and within either adjacent pair every degree unequal to a contributes the same signed unit, the number of internal degrees unequal to a is $O(m)$. Externally, the coverage hypothesis and the incidence sum give

$$\sum_{Q \notin \mathcal{K}} (\sigma_n(Q) - \lambda) = (q + 1 - n)T - \lambda b = O(m),$$

so only $O(m)$ external degrees differ from λ . Thus only $O(m) = O(q)$ dual lines have the wrong intersection residue.

For all sufficiently large q , this number lies below the Szőnyi–Weiner threshold quoted above. Their theorem changes $\mathcal{L}^*$ at exactly $\lceil \delta/(q+1) \rceil = O(1)$ distinct support points to obtain the required $\lambda \pmod{p}$ multiset. We take the support modulo p , deleting any point whose correction coefficient is zero; every listed support point therefore has nonzero coefficient modulo p . $\square$

The following observation is where completeness re-enters the modular argument.

Lemma 5.2 (Contribution of the repair support). *Fix an ordered factorization $\lambda = uv$ and suppose $d = u + v + 1$ is a power of the characteristic. Let*

$$q = dm, \quad n = (u+1)m + 1, \quad k = udm^2 + cm + o(m), \quad T = ud(v+1)m + h + o(1),$$

where $h \in \mathbb{Z}$. Suppose every external point is on at least λ n -secants, and suppose $\mathcal{L}^*$ can be changed at r distinct support points, with nonzero correction coefficient modulo p at each, into a multiset meeting every line in $\lambda \pmod{p}$ points. If r is fixed and nonzero, then

$$c \geq L(h) + \frac{r}{u}. \quad (5.1)$$

Proof. Dualize the r support points to r lines of the original plane. Away from their pairwise intersections, a primal point on one support line corresponds dually to a line containing exactly one support point. The correction coefficient there is nonzero modulo p , so the original n -secant multiplicity differs from $\lambda \pmod{p}$. If the point is external, its multiplicity is at least $\lambda + 1$. A support line contains at most n points of $\mathcal{K}$, and the union loses only $O(r^2)$ points at pairwise intersections. Hence the total external excess over multiplicity λ is at least

$$r(q + 1 - n) - O(r^2) = rvm - O(r^2). \quad (5.2)$$

On the other hand, the incidence sum gives the exact first-order expression

$$(q + 1 - n)T - \lambda(q^2 + q + 1 - k) = uv(c - L(h))m + o(m). \quad (5.3)$$

Comparing (5.2) and (5.3) proves (5.1). $\square$

There is a useful restriction when $r = 1$. Summing the line intersections of an exact $\lambda \pmod{p}$ multiset over a pencil shows that its total multiplicity is $\lambda \pmod{p}$. Along a characteristic-compatible family write $d = p^j$ and $q = p^e = dm$. Then $m = p^{e-j}$ is divisible by p once $e > j$, so the leading term of T is divisible by p . The correction therefore has total multiplicity $\lambda - h \pmod{p}$. Consequently, if

$$h \equiv \lambda \pmod{p}, \quad (5.4)$$

then a nonzero correction cannot have support one.

We apply this first to ordinary completeness. Write $q = 3m$, $n = 2m + 1$, and suppose

$$k = 3m^2 + cm + o(m), \quad T = 6m + h + o(1). \quad (5.5)$$

The coverage and pair-count inequalities from Sections 3 and 4 become

$$c + h \geq 3, \quad 5c - h \geq 18. \quad (5.6)$$

Lemma 5.3 (The uncorrected case is impossible). *Let $\mathcal{K}$ be a complete (k, n) -arc with parameters (5.5), and let $\mathcal{L}^*$ be the dual point set of all its n -secants. If $c < 4$ and every line meets $\mathcal{L}^*$ in $1 \pmod{3}$ points, then the exact pair identity (3.3) cannot hold.*

Proof. Near this equality parameter the balanced internal multiplicities are 3, 4, and the balanced external multiplicities are 1, 2. Let D be the signed internal deficiency $4k - nT$, and let E be the total external excess above one. The incidence sums give

$$D = (4c - 2h - 6)m + o(m), \quad (5.7)$$

$$E = (c + h - 3)m + o(m). \quad (5.8)$$

Completeness gives $E \geq 0$. Moreover, (5.6) and $c < 4$ give $D \geq (30 - 6c)m + o(m) > 0$. Without the congruence restriction, the available pair-count slack is

$$(5c - h - 18)m + o(m). \quad (5.9)$$

If every multiplicity is $1 \pmod{3}$, congruence-restricted integer convexity replaces the internal adjacent values 3, 4 by 1, 4, raising the minimum pair count by D . It replaces the external adjacent values 1, 2 by 1, 4, raising that minimum by E . Feasibility would therefore require

$$5c - h - 18 \geq \frac{D + E}{m} + o(1) = 5c - h - 9 + o(1),$$

which is impossible. $\square$

Proposition 5.4 (Characteristic-three modular-repair bound). *For every $\varepsilon > 0$, all sufficiently large powers $q = 3^e$ have the following property: every complete $(k, 2q/3 + 1)$ -arc in $\text{PG}(2, q)$ satisfies*

$$k \geq \frac{q^2}{3} + \left(\frac{4}{3} - \varepsilon\right)q.$$

Proof. Suppose the assertion fails. For some $\varepsilon > 0$, pass to a sequence with

$$k \leq 3m^2 + (4 - 3\varepsilon)m.$$

The classical incidence bound supplies a lower bound for $c_m = (k - 3m^2)/m$, so, after passing to a subsequence, $c_m \rightarrow c$ with $c \leq 4 - 3\varepsilon$. The leading real inequality has a unique equality point at $T/m = 6$. Lemma 4.2 then gives $T - 6m = O(1)$; pass to a further subsequence on which this difference is a fixed integer h .

For the actual degree sequences, the pair identity gives the exact decomposition

$$-F(T) = \left(\sum_{P \in \mathcal{K}} \binom{\rho_n(P)}{2} - \Phi_{\min}(k, nT) \right) + \left(\sum_{Q \notin \mathcal{K}} \binom{\sigma_n(Q)}{2} - \Phi_{\min}(b, (q + 1 - n)T) \right).$$

Both parentheses are nonnegative. Equation (4.13), with bounded c_m and fixed h , shows that their sum is $O(m)$. Equation (2.3) therefore shows that only $O(m)$ internal degrees lie outside $\{3, 4\}$, and only $O(m)$ external degrees lie outside $\{1, 2\}$.

It remains to count the wrong adjacent degrees. The signed deficiency $D = 4k - nT = O(m)$ equals $\sum_{P \in \mathcal{K}} (4 - \rho_n(P))$. For every nonnegative integer $z \notin \{3, 4\}$,

$$|4 - z| \leq (z - 3)(z - 4),$$

so the total contribution of the nonadjacent internal degrees is $O(m)$ by (2.3). Hence the number of internal degree-three points is $O(m)$. Externally, completeness gives $\sigma_n(Q) \geq 1$, and the excess $E = \sum_Q (\sigma_n(Q) - 1) = O(m)$ bounds the number of degree-two points. Thus only $O(m) = O(q)$ dual lines fail to meet $\mathcal{L}^*$ in 1 (mod 3) points. Szőnyi–Weiner’s repair theorem changes $\mathcal{L}^*$ at a bounded integer number r of support points to obtain an exact 1 (mod 3) multiset. Pass to a further subsequence on which r is fixed. Lemma 5.3 excludes $r = 0$. Lemma 5.2 gives

$$c \geq L(h) + r = 3 - h + r.$$

By (5.4), $r \geq 2$ when $h \equiv 1 \pmod{3}$, and $r \geq 1$ otherwise. Together with the pair-count bound $c \geq (18 + h)/5$, this yields

$$c \geq \min_{h \in \mathbb{Z}} \max \left\{ \frac{18 + h}{5}, 3 - h + r_{\min}(h) \right\} = 4, \quad (5.10)$$

where $r_{\min}(h) = 2$ for $h \equiv 1 \pmod{3}$ and is one otherwise. This contradicts the chosen sequence and proves the proposition. $\square$

6 A ternary line-code obstruction

We now sharpen Proposition 5.4. The preceding repair argument approaches the center

$$k_0(q) = \frac{q^2}{3} + \frac{4q}{3}.$$

At this center, the first two secant moments produce a small-weight word in the ternary code generated by the lines of $\text{PG}(2, q)$. Two pointwise integer-shell inequalities force that word to need at least three generator lines and show that three lines already cost an additional $q/3 - o(q)$ points.

Fix $q = 3r$, put $n = 2r + 1$, and let $\mathcal{A}$ be a complete (K, n) -arc. Let $\mathcal{D}$ be the point set in the dual plane corresponding to the full family of n -secants. After fixing a polarity, write M for the symmetric point–line incidence matrix and

$$d = M\mathbf{1}_{\mathcal{D}}, \quad T = |\mathcal{D}|.$$

Thus d_P is the number of n -secants through P , and completeness gives $d_P \geq 1$ for $P \notin \mathcal{A}$. The incidence and pair identities are

$$\sum_{P \in \mathcal{A}} d_P = nT, \quad \sum_{P \notin \mathcal{A}} d_P = rT, \quad \sum_P \binom{d_P}{2} = \binom{T}{2}. \quad (6.1)$$

Write

$$K = 3r^2 + 4r + \delta, \quad T = 6r + j, \quad u = \mathbf{1} + 3\mathbf{1}_{\mathcal{A}} - d. \quad (6.2)$$

Lemma 6.1 (Centered moment identities). *With the notation above,*

$$\sum_P u_P = (9 - 3j)r + 3\delta - j + 1, \quad (6.3)$$

$$\sum_P u_P^2 = (15 - 3j)r + 15\delta + j^2 - 8j + 1, \quad (6.4)$$

$$\sum_{P \in \mathcal{A}} u_P = (10 - 2j)r + 4\delta - j. \quad (6.5)$$

Moreover, the shell defect

$$E = \sum_{P \in \mathcal{A}} \frac{u_P(u_P - 1)}{2} + \sum_{P \notin \mathcal{A}} \frac{u_P(u_P + 1)}{2} \quad (6.6)$$

is nonnegative and equals

$$E = (2 - j)r + 1 + 5\delta + \frac{j(j - 7)}{2}. \quad (6.7)$$

Proof. The first and third identities follow from $\sum_P d_P = (q + 1)T$ and the first identity in (6.1). The pair identity gives $\sum_P d_P^2 = T(T + q)$; expanding u^2 gives (6.4). Expanding (6.6) and applying the same moments gives (6.7). Every shell summand is nonnegative: on the complement, completeness gives $u_P = 1 - d_P \leq 0$. $\square$

Let $C_1(2, q)$ denote the ternary code generated by the line incidence vectors. Reducing u modulo three gives

$$\bar{u} = \mathbf{1} - M\mathbf{1}_{\mathcal{D}} \in C_1(2, q), \quad (6.8)$$

because the sum of the lines in a pencil is $\mathbf{1}$ modulo three. The identity $M^2 = qI + J$ also gives

$$Mu = (q + 1 - T)\mathbf{1} + 3M\mathbf{1}_{\mathcal{A}} - q\mathbf{1}_{\mathcal{D}}. \quad (6.9)$$

We use the following exact form of the small-codeword theorem. If a nonzero word $c \in C_1(2, q)$, where $q = p^e > 27$ and $e > 2$, has weight

$$w(c) < (\lfloor \sqrt{q} \rfloor + 1)(q + 1 - \lfloor \sqrt{q} \rfloor),$$

then Szőnyi–Weiner [18, Theorem 4.3] express c as a linear combination of exactly $\lceil w(c)/(q + 1) \rceil$ distinct lines. In characteristic three its nonzero coefficients have balanced representatives in $\{-1, 1\}$.

Theorem 6.2 (Three-line gap). *Fix $C > 0$ and $\varepsilon > 0$. For all sufficiently large $q = 3^e$, there is no complete $(K, 2q/3 + 1)$ -arc in $\text{PG}(2, q)$ with*

$$K = \frac{q^2}{3} + \frac{4q}{3} + \delta, \quad -Cq \leq \delta \leq \left(\frac{1}{3} - \varepsilon\right)q. \quad (6.10)$$

Proof. Suppose otherwise, and write $\delta = \alpha q$. Applying Cauchy–Schwarz separately to the two classes in (6.1) gives

$$F(T) := \frac{n^2 T^2}{K} - nT + \frac{r^2 T^2}{q^2 + q + 1 - K} - rT - T(T - 1) \leq 0. \quad (6.11)$$

Coverage gives $T \geq (q^2 + q + 1 - K)/r$, so it bounds $j = T - 6r$ below in terms of C . Write $F(T) = AT^2 - 3rT$, with

$$A = \frac{n^2}{K} + \frac{r^2}{q^2 + q + 1 - K} - 1,$$

which is positive by the same weighted Cauchy argument as in Section 3. Since $F(T) \leq 0$ and $T > 0$, $T \leq 3r/A$. Uniformly for α in the compact interval prescribed by (6.10), direct expansion gives

$$\frac{3r}{A} = 6r + 5 + 15\alpha + O_{C, \varepsilon}(r^{-1}).$$

Thus j is bounded above as well. Pass to a subsequence on which j is fixed. For later reference, direct expansion now gives

$$F(6r + j) = 3(j - 5 - 15\alpha)r + O_{C,\varepsilon,j}(1).$$

The centered identities then give $E = O(q)$ and $\sum_P u_P^2 = O(q)$.

Let $z_P \in \{-1, 0, 1\}$ be the balanced representative of $u_P \pmod{3}$, put $R = \{P : z_P \neq 0\}$, and write $w = |R|$. For the summand e_P of (6.6), the pointwise inequality

$$2\mathbf{1}_{\mathcal{A}}(P)u_P - e_P - u_P \leq \mathbf{1}_R(P) \quad (6.12)$$

holds. On the arc its left side is $u_P - u_P(u_P - 1)/2$, while off the arc it is $-u_P(u_P + 3)/2$; the three residue classes prove the claim. Summing and using Lemma 6.1 gives the exact cancellation

$$3q - \frac{(j-1)(j-4)}{2} = 2 \sum_{P \in \mathcal{A}} u_P - E - \sum_P u_P \leq w. \quad (6.13)$$

By (6.8), $\bar{u}$ belongs to the ternary line code, and $w \leq \sum_P u_P^2 = O(q)$. Szőnyi–Weiner’s theorem therefore gives

$$\bar{u} = \sum_{i=1}^t \varepsilon_i \mathbf{1}_{L_i}, \quad t = \left\lceil \frac{w}{q+1} \right\rceil, \quad \varepsilon_i \in \{-1, 1\},$$

on distinct lines L_i . The norm bound makes t bounded. Away from their $O(t^2)$ intersection points, every generator line contributes a nonzero coordinate, so

$$w = tq + O(1). \quad (6.14)$$

Equation (6.13) forces

$$t \geq 3. \quad (6.15)$$

Put $\sigma = \sum_i \varepsilon_i$, and let $a = (t + \sigma)/2$ be the number of positive generator lines. Pointwise,

$$(u_P^2 - z_P^2) - (u_P - z_P) \geq 0. \quad (6.16)$$

Indeed $u_P \equiv z_P \pmod{3}$; if $z_P = 1$ and $P \notin \mathcal{A}$, completeness gives $u_P \leq -2$, and the left side is at least six. The positive generator lines contain $aq + O(1)$ nonintersection points with $z = 1$, but at most an of them lie on the arc. Summing (6.16) consequently gives

$$\left(\sum_P u_P^2 - w \right) - \left(\sum_P u_P - \sum_P z_P \right) \geq 2aq - O(1). \quad (6.17)$$

Since $\sum_P z_P = \sigma q + O(1)$, substitution of (6.3), (6.4), and (6.14) into (6.17) cancels j and σ , leaving

$$t \leq 1 + 6\alpha + o(1). \quad (6.18)$$

Together with (6.15), this gives $\alpha \geq 1/3 - o(1)$, contrary to $\alpha \leq 1/3 - \varepsilon$. $\square$

Proposition 6.3 (Sublinear-repair rigidity). *Suppose that a sequence of complete $(K, 2q/3 + 1)$ -arcs over $q = 3^e$ satisfies*

$$K = \frac{q^2}{3} + \frac{5q}{3} + o(q).$$

Then, for every sufficiently large member of the sequence, the centered residue word $\bar{u}$ is an exact signed combination of three distinct lines.

Proof. Coverage and (6.11) again keep $j = T - 2q$ bounded. Equations (6.15) and (6.18), now with $\alpha = 1/3 + o(1)$, give $3 \leq t \leq 3 + o(1)$. Since t is an integer, it equals three for all sufficiently large fields. The exact small-codeword theorem supplies the claimed signed representation. $\square$

Proposition 6.4 (Exact endpoint obstruction). *For all sufficiently large $q = 3^e$, there is no complete*

$$\left(\frac{q^2}{3} + \frac{5q}{3}, \frac{2q}{3} + 1 \right) \text{-arc in PG}(2, q).$$

Proof. Suppose that such arcs exist for arbitrarily large q , and pass to a sequence of them. Put $q = 3r$ and $\delta = r$ in (6.2). Proposition 6.3 shows that, for every sufficiently large member of the sequence,

$$z = \varepsilon_1 \mathbf{1}_{L_1} + \varepsilon_2 \mathbf{1}_{L_2} + \varepsilon_3 \mathbf{1}_{L_3} \pmod{3}, \quad \varepsilon_i \in \{-1, 1\},$$

where $z_P \in \{-1, 0, 1\}$ is the balanced representative of u_P . Write $\sigma = \varepsilon_1 + \varepsilon_2 + \varepsilon_3$.

We first record the exact scalar reduction. Put $R_+ = \{P : z_P = 1\}$, $R_- = \{P : z_P = -1\}$, and let $a = (3 + \sigma)/2$ be the number of positive generator lines. Pointwise on $\mathcal{A}$, one has $u_P \leq e_P + \mathbf{1}_{R_+ \cup R_-}(P)$; off $\mathcal{A}$, one has $e_P \geq 1$ on R_+ . Therefore

$$\sum_{P \in \mathcal{A}} u_P \leq E - |R_+| + 2|\mathcal{A} \cap R_+| + |\mathcal{A} \cap R_-|. \quad (6.19)$$

The signed three-line representation gives $|R_+| = aq + O(1)$. The arc cap on its three generator lines gives

$$|\mathcal{A} \cap R_+| \leq a(2q/3 + 1) + O(1), \quad |\mathcal{A} \cap R_-| \leq (3 - a)(2q/3 + 1) + O(1).$$

Substitution of the exact centered identities in (6.19) yields $j \geq 1 + a$ for all sufficiently large q .

Also $E \geq 0$ gives $j \leq 7$, while summing the residue word and comparing u with its balanced representative give

$$\sigma \equiv 1 - j \pmod{3}, \quad |4 - j - \sigma| \leq 7 - j. \quad (6.20)$$

Indeed the second inequality is the coefficient of q in $|\sum_P (u_P - z_P)| \leq \sum_P (u_P^2 - z_P^2)$. Checking the four values $\sigma \in \{-3, -1, 1, 3\}$ under $1 + a \leq j \leq 7$ and (6.20) leaves

$$(j, \sigma) = (1, -3), (2, -1), (3, 1), (4, -3), (4, 3), (5, -1), (7, -3).$$

For the last row, $\sum_P u_P^2 = 3q - 6 < w$, so the six possibilities are

$$(j, \sigma) = (1, -3), (2, -1), (3, 1), (4, -3), (4, 3), (5, -1). \quad (6.21)$$

The three lines cannot be concurrent. When their signs agree, the exact inequality above gives the following contradictions; the two numerical columns are $|\sum u - \sum z|$ and $\sum u^2 - w$, respectively:

(j, σ)	$ \sum u - \sum z $	$\sum u^2 - w$
$(1, -3)$	$6q$	$6q - 6$
$(4, -3)$	$3q - 3$	$3q - 15$
$(4, 3)$	$3q + 3$	$3q - 15$

For the mixed-sign rows, the arc cap leaves at least $q/3 - 1$ external positive nonvertices when $j = 2, 5$, and at least $2q/3 - 1$ external positive residues when $j = 3$. Substitution in the exact pointwise correction (6.16) makes (6.17) fail by six in each case.

The generators therefore form a triangle. Its residue support has size $3q$ when the signs agree and $3q - 2$ when they are mixed. Comparing with (6.13) immediately excludes the rows $(2, -1)$ and $(3, 1)$. In each remaining row equality holds in the pointwise inequality (6.12). Checking its three residue classes gives the exact degree rule

$$z_P = 1 \implies d_P = 3, \quad z_P = -1 \implies d_P = 2, \quad z_P = 0 \implies d_P \in \{1, 4\}. \quad (6.22)$$

Dualize the generator lines to noncollinear points Q_1, Q_2, Q_3 , put $p_i = (1 + \varepsilon_i)/2$, and put $y_i = \mathbf{1}_{\mathcal{D}}(Q_i)$. If x_{ij} is the $\mathcal{D}$ -degree of the connector $Q_i Q_j$, then the $q - 1$ ordinary lines through Q_i have degree $2 + p_i$ by (6.22). Counting incidences with $\mathcal{D}$ in that pencil gives the general identity

$$x_{ij} + x_{ik} = j + 2 + p_i + q(y_i - p_i), \quad \{i, j, k\} = \{1, 2, 3\}. \quad (6.23)$$

For $(j, \sigma) = (1, -3)$, all three connectors have degree three, so (6.23) would give $6 = 3 + qy_i$, impossible for $q > 3$. For $(4, 3)$, all connectors have degree two, and the same equation gives $4 = 7 + q(y_i - 1)$, again impossible for $q > 3$. For $(4, -3)$, it gives $y_i = 0$ and connector degrees $(3, 3, 3)$. Finally take $(5, -1)$, with one positive generator. At its point Q_i , the two connectors have degrees in $\{1, 4\}$, and (6.23) forces $y_i = 1$ and their sum to be eight; both therefore have degree four. At either negative generator the remaining connector has degree three, so the same equation forces $y_i = 0$. Thus the only surviving rows and connector profiles are

$$(j, \sigma) = (4, -3), (5, -1), \quad (x_{12}, x_{13}, x_{23}) = (3, 3, 3), (4, 4, 3),$$

up to relabeling.

For $(j, \sigma) = (4, -3)$, the positive residue consists only of the three triangle vertices. On $\mathcal{A}$, the pointwise inequality $u_P \leq 2e_P + \mathbf{1}_{\{z_P=1\}}$ therefore gives

$$2q - 4 = \sum_{P \in \mathcal{A}} u_P \leq 2E + 3 = 2q - 7,$$

a contradiction.

It remains to exclude $(j, \sigma) = (5, -1)$. Let N_i be the number of lines of $\mathcal{D}$ -degree i . The equality rule (6.22) gives $N_2 = 2q - 2$ and $N_3 = q$, while the total-line and first-moment identities give

$$N_4 = \frac{q(q+2)}{3} + 2.$$

Let A_i count points of $\mathcal{A}$ whose dual lines have $\mathcal{D}$ -degree i , put $b = A_1$, and put $h = N_4 - A_4$. The selected-point and selected-incidence equations reduce to

$$A_3 = \frac{2q}{3} + 1 + b + 2h.$$

Only one degree-three point lies off the positive generator line, so its arc cap gives $b + 2h \leq 1$. Hence $h = 0$, and both degree-four connector vertices are selected. At least $2q/3$ selected degree-three points then lie on the positive generator; the two connector vertices raise its intersection with $\mathcal{A}$ to at least $2q/3 + 2$, contradicting the maximum $2q/3 + 1$. This excludes the last row. $\square$

Proof of Theorem 1.1. Fix $\varepsilon > 0$. If $\varepsilon > 1/3$, Proposition 5.4 already gives the claim, so assume $0 < \varepsilon \leq 1/3$. If the theorem failed along an unbounded sequence, that proposition would give

$$K \geq \frac{q^2}{3} + \frac{4q}{3} - o(q),$$

whereas failure would give

$$K \leq \frac{q^2}{3} + \frac{4q}{3} + \left(\frac{1}{3} - \varepsilon\right)q.$$

For large q , these inequalities put δ in the range of Theorem 6.2, with any fixed $C > 0$, a contradiction. $\square$

7 The characteristic-two family

We turn to the ordered factorization $(u, v) = (2, 1)$. Thus

$$q = 8m, \quad n = 6m + 1, \quad \lambda = 2.$$

Near equality write

$$k = 32m^2 + cm + o(m), \quad T = 32m + h + o(1). \quad (7.1)$$

The integer pair condition and the external incidence count give

$$6c - h \geq 76, \quad e = 2(c + h - 8)m + o(m), \quad (7.2)$$

where $e = \sum_{Q \notin \mathcal{K}} (\sigma_n(Q) - 2)$ is the total excess above the required two n -secants.

The dual set $\mathcal{L}^*$ has only $O(q)$ odd-secants. For sufficiently large even q , this is below the Szőnyi–Weiner threshold $(\lfloor \sqrt{q} \rfloor + 1)(q + 1 - \lfloor \sqrt{q} \rfloor)$. Their stability theorem [19, Theorem 1.1] therefore writes it as the symmetric difference

$$\mathcal{L}^* = \mathcal{E} \triangle R, \quad (7.3)$$

where every line meets $\mathcal{E}$ in an even number of points and $|R| = r = \lceil \delta/(q + 1) \rceil = O(1)$, with δ the number of odd-secants of $\mathcal{L}^*$.

Lemma 7.1 (The even-type case). *Let $\mathcal{K}$ be a (k, n) -arc with parameters (7.1), suppose every external point lies on at least two n -secants, and let $\mathcal{L}^*$ be the dual point set of all its n -secants. If $R = \emptyset$ in (7.3), then $c \geq 16$.*

Proof. The balanced external pair is $\{2, 3\}$. Put $D = 6k - nT$, the signed internal deficiency from six. The incidence equations and the unrestricted integer pair bound give

$$\begin{aligned} D &= (6c - 6h - 32)m + o(m), \\ e &= 2(c + h - 8)m + o(m), \\ S_{\text{aff}} &= (12c - 2h - 152)m + o(m). \end{aligned}$$

If $D \geq 0$, the balanced internal pair is $\{5, 6\}$, and S_{aff} is the available unrestricted pair slack to first order. If $D < 0$, the balanced pair is $\{6, 7\}$; convexity of the internal $\Phi_{\min}$ term makes the same low-branch affine continuation S_{aff} an upper bound on the actual unrestricted slack. If every multiplicity is even, restricting the internal balanced distribution to even integers raises its minimum by $|D|/2$, while the external restriction raises its minimum by $e/2$. The total increase is $(|D| + e)/2$. Since $|D| \geq D$, it is at least

$$\frac{D + e}{2} = (4c - 2h - 24)m + o(m).$$

In either branch, feasibility forces S_{aff} to be at least this large, which simplifies to $c \geq 16$. $\square$

Lemma 7.2 (Parity of the repair). *Suppose $R \neq \emptyset$ in (7.3). If r is odd, then*

$$c + h \geq 9. \quad (7.4)$$

If r is even, then

$$c + h \geq 10. \quad (7.5)$$

Moreover, $r \equiv T \equiv h \pmod{2}$.

Proof. Every even-type set has even cardinality, so $r \equiv |\mathcal{L}^*| = T \pmod{2}$; the leading term $32m$ is even, giving the last congruence.

Dualize the points of R to r lines of the original plane. On each such line, at most $r - 1$ points can cancel with the other generators in the symmetric difference. Thus at least $q - r + 2$ points have odd n -secant multiplicity. At most $n = 3q/4 + 1$ of them lie in $\mathcal{K}$, so a single generator contributes at least $q/4 - r + 1$ external points to the excess e .

If r is odd, then $r \geq 1$ and hence $e \geq q/4 - O(1) = 2m - O(1)$. If r is even, then $r \geq 2$; taking the union of two generator lines loses only their intersection, and $e \geq q/2 - O(1) = 4m - O(1)$. Comparing with (7.2) proves (7.4) and (7.5). $\square$

Proof of Theorem 1.3. Suppose the theorem fails. Pass to a sequence for which the coefficient $c_m = (k - 32m^2)/m$ is bounded above by $73/6 - 8\varepsilon$. The classical incidence bound bounds c_m below, so pass to a subsequence with $c_m \rightarrow c$. The leading incidence bound and Lemma 4.2 give $T - 32m = O(1)$. Pass to a further subsequence with fixed integer difference h .

Equation (7.2) and $c \leq 73/6$ give

$$6c - 6h - 32 \geq 424 - 30c \geq 59.$$

Thus $D > 0$ for all sufficiently large m , so the balanced internal pair in this contradiction range is $\{5, 6\}$.

As in the characteristic-three proof, the exact decomposition of $-F(T)$ into the two nonnegative balancing losses and Equation (2.3) show that only $O(m)$ internal degrees lie outside $\{5, 6\}$, and only $O(m)$ external degrees lie outside $\{2, 3\}$. Moreover,

$$D = 6k - nT = \sum_{P \in \mathcal{K}} (6 - \rho_n(P)) = O(m).$$

For $z \notin \{5, 6\}$, the absolute value of $6 - z$ is at most twice the corresponding summand in (2.3); hence the number of internal degree-five points is $O(m)$. Since every external degree is at least two and $e = \sum_Q (\sigma_n(Q) - 2) = O(m)$, the number of external degree-three points is also $O(m)$. These are the wrong-parity adjacent degrees, so only $O(m) = O(q)$ dual lines have the wrong parity. Therefore Szőnyi–Weiner’s theorem gives (7.3) with bounded integer r . Pass to a further subsequence on which r is fixed. Lemma 7.1 excludes $r = 0$ in the range under consideration. If h is odd, Lemma 7.2 and (7.2) give

$$c \geq \max \left\{ \frac{76 + h}{6}, 9 - h \right\}.$$

The minimum over odd h is $73/6$, attained at $h = -3$. If h is even, the corresponding bound is

$$c \geq \max \left\{ \frac{76 + h}{6}, 10 - h \right\},$$

whose minimum over even h is $37/3$, attained at $h = -2$. Both contradict the chosen sequence. Since $q = 8m$, this proves (1.7). $\square$

The parity argument uses only the repair theorem and the fact that an n -secant contains at most n points of the arc. It does not require a classification of even-type sets of size $4q + O(1)$.

8 Blocking sets and projective codes

The complement of a (k, n) -arc $\mathcal{K}$ in a projective plane of order q is a $(q + 1 - n)$ -fold blocking set. Completeness says more: each point of the complement lies on a line meeting it in exactly $q + 1 - n$ points. Thus the complement is a minimal multiple blocking set. Under this correspondence, n -secants of $\mathcal{K}$ are the $(q + 1 - n)$ -secants of its complement, and the hypothesis $\sigma_n(Q) \geq \lambda$ says that every point of the blocking set is on at least λ such secants. This is precisely the dual setting of Bishnoi–Mattheus–Schillewaert [8, Section 8]; compare also Ball’s finite-plane treatment [3].

In particular, Theorem 1.1 is equivalent to the following statement.

Corollary 8.1. *Let $q = 3^h$ and let B be the complement of a complete $(k, 2q/3 + 1)$ -arc in $\text{PG}(2, q)$. Then B is a minimal $q/3$ -fold blocking set and*

$$|B| \leq \frac{2q^2}{3} - \frac{2q}{3} + 1 + o(q).$$

There is also a standard coding interpretation; compare Kohnert’s incidence criterion for projective code extension [12] and the multiple-blocking-set/code correspondence of Ball–Fancsali [4]. If the (k, n) -arc spans the plane, projective representatives of its points form the columns of a generator matrix for a projective $[k, 3, k - n]_q$ code. A line meeting the arc in n points corresponds to a minimum-weight projective codeword. Adding a new projective column raises the minimum distance by one exactly when no minimum-weight hyperplane contains that column. Hence completeness is equivalent to nonextendibility. The condition $\sigma_n(Q) \geq \lambda$ records at least λ minimum-weight hyperplanes through every candidate column.

The same statement is available in every projective dimension $r - 1 \geq 2$. Points and hyperplanes of $\text{PG}(r - 1, q)$ form a symmetric design with parameters

$$v = \frac{q^r - 1}{q - 1}, \quad K = \frac{q^{r-1} - 1}{q - 1}, \quad \Lambda = \frac{q^{r-2} - 1}{q - 1}. \quad (8.1)$$

Corollary 8.2 (Projective codes). *Let C be a projective $[k, r, k - n]_q$ code, and let $\mathcal{S}$ be the hyperplanes containing n columns of a projective system for C . Partition the points of $\text{PG}(r - 1, q)$ into the k columns of C and the remaining candidate columns. For every integer $T = |\mathcal{S}|$, the two selected-hyperplane degree sequences satisfy the exact interval condition (2.4), with the parameters in (8.1). Any prescribed lower bound on the number of hyperplanes in $\mathcal{S}$ through a candidate column may be used as the lower endpoint of its integer degree interval.*

Proof. Apply Theorem 2.2 to the point–hyperplane design of $\text{PG}(r - 1, q)$. A minimum-weight hyperplane contains exactly n columns, so the selected blocks have constant intersection size n with the first point class. $\square$

Exact two-valued hyperplane intersections correspond to projective two-weight codes [9]. Consider a near-equality sequence satisfying the hypotheses of Theorem 1.2, with bounded first-order coefficient and with the coverage and pair-count bounds simultaneously tight to first order at an integral h_* from Proposition 4.3. The balancing-loss argument in the proofs of Proposition 5.4 and Theorem 1.3 then shows that all but $O(q)$ dual lines have the two intersection numbers

$$uv \quad \text{and} \quad (u + 1)(v + 1).$$

Their difference is $u + v + 1$, which is a power of the characteristic in the parameter family considered here: $u + v + 1$ divides $q = p^e$ because $q = (u + v + 1)m$. Thus the two limiting

values are congruent modulo p . The present argument does not classify the $O(q)$ exceptional lines.

The exact two-character case does, however, admit a canonical inverse. The following observation is the dual incidence form of the classical two-character-set dictionary [9].

Proposition 8.3 (Exact two-character inverse). *Let B be a nonempty proper point set in the dual of $\text{PG}(2, q)$, and suppose both line characters $\lambda < a$ occur, with $\lambda \geq 1$. Let $\mathcal{S}$ be the primal lines dual to the points of B , put $d = a - \lambda$, and define*

$$\mathcal{A} = \{P : \deg_{\mathcal{S}}(P) = a\}.$$

Then $d \mid q$, and $\mathcal{A}$ is a complete (k, n) -arc whose full set of n -secants is $\mathcal{S}$, where

$$k = \frac{|B|(q+1) - \lambda(q^2 + q + 1)}{d}, \quad n = \frac{|B| + q - \lambda(q+1)}{d}.$$

Every external point of $\mathcal{A}$ lies on exactly λ n -secants, and every line outside $\mathcal{S}$ meets $\mathcal{A}$ in $n - q/d$ points.

Proof. For a dual point x , let $N_a(x)$ be the number of lines through x that meet B in a points. Summing $|L \cap B|$ over the dual lines through x gives $|B| + q$ when $x \in B$, because x is counted on all $q + 1$ lines, and gives $|B|$ when $x \notin B$. Therefore

$$N_a(x) = \begin{cases} (|B| + q - \lambda(q+1))/d, & x \in B, \\ (|B| - \lambda(q+1))/d, & x \notin B. \end{cases}$$

The two values differ by q/d , so $d \mid q$. Under duality, $N_a(\ell^*) = |\ell \cap \mathcal{A}|$. Thus the lines in $\mathcal{S}$ meet $\mathcal{A}$ in n points and every other line meets it in $n - q/d < n$ points. Hence $\mathcal{S}$ is exactly the full set of maximal secants. An external point has $\mathcal{S}$ -degree λ , so $\lambda \geq 1$ proves completeness and the stated multiplicity.

Finally, summing $|L \cap B|$ over all dual lines gives

$$|B|(q+1) = ak + \lambda(q^2 + q + 1 - k),$$

which yields the formula for k . □

9 Concluding remarks

The first two incidence moments already contain the classical spectral bound. Their sharp integer degree ranges retain information that the real relaxation loses, and on each rational equality family classified in Theorem 1.2 this loss has linear order. Modular stability then converts the nearly two-valued dual intersection distribution into a bounded repair. Completeness assigns a further linear cost to every point in that repair support.

Proposition 5.1 gives a near-extremal structural reduction. For every compatible characteristic, at the leading equality density the dual set of all maximal secants differs at only $O(1)$ support points from an exact $\lambda \pmod{p}$ multiset. The proposition does not classify those exact modular cores or determine which bounded perturbations are geometrically realizable.

Proposition 8.3 closes the inverse-realization problem for an exact two-character dual set: the high-character lines recover the arc canonically, and the dual point set is exactly its maximal-secant family. The remaining inverse problem begins only after bounded repair, where the perturbed modular core need not retain exact maximal intersections.

The line-code obstruction shows that the modular-repair coefficient $4/3$ is not asymptotically sharp and raises the lower coefficient to $5/3$. We do not prove that the new coefficient is optimal. The remaining extremal problem is to construct complete arcs on this scale or to find further obstructions that raise the coefficient again. A second problem is structural: classify the exact modular cores left by Proposition 5.1 and determine which bounded perturbations can occur as the full set of n -secants of a complete (k, n) -arc.

Proposition 6.3 narrows the construction problem: every asymptotically matching family must have a centered residue word generated by exactly three lines. This is a necessary structural condition, not a construction or a classification of the realizable three-line cores.

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This project was developed with extensive assistance from OpenAI Codex and Anthropic Claude in proof exploration, literature research, exact computation, adversarial review, and drafting. The author checked the arguments, computations, code, and cited sources, reviewed and edited all AI-assisted material, and assumes responsibility for all content.

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